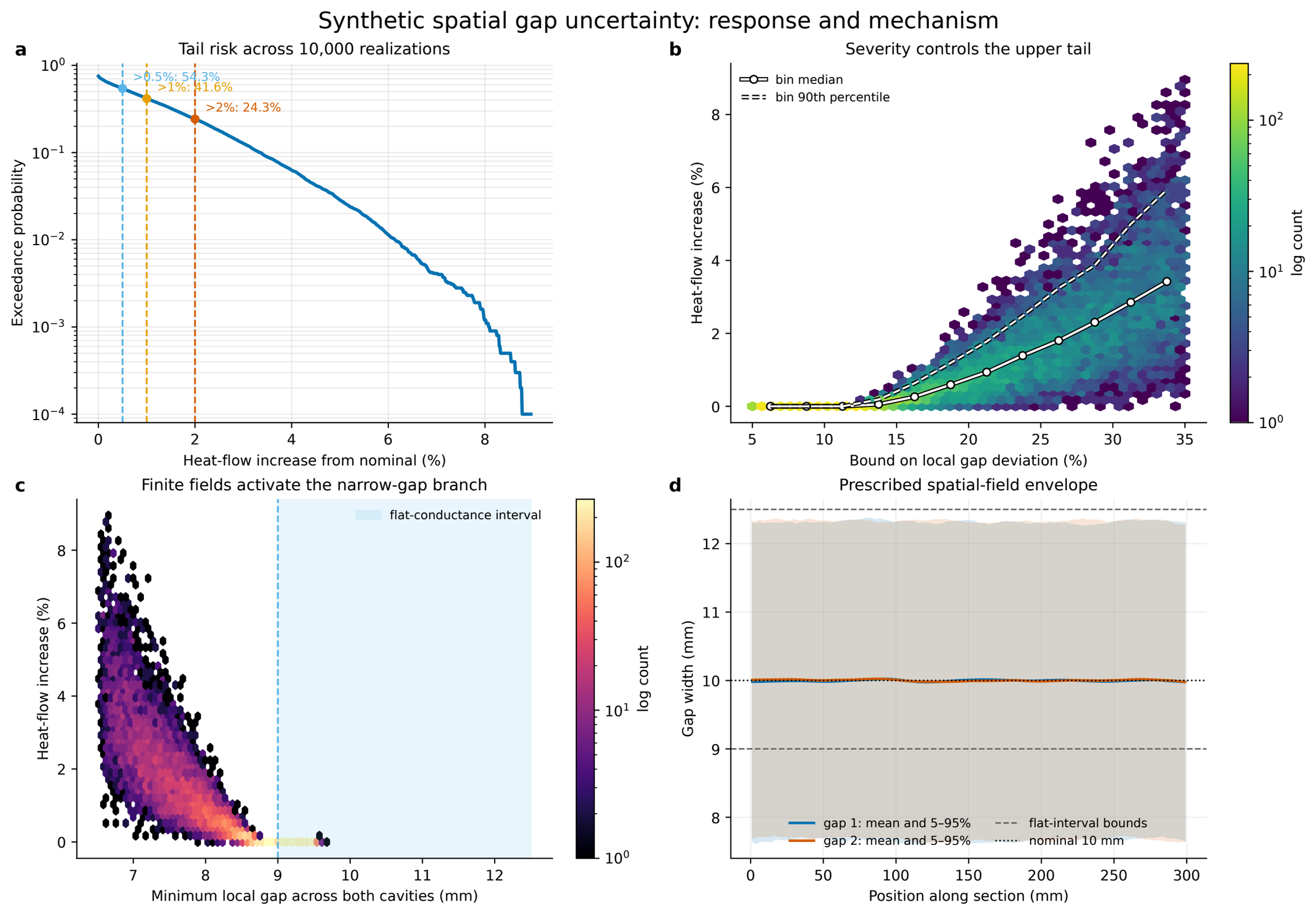
Synthetic spatial gap uncertainty: response and mechanism
a
Tail risk across 10,000 realizations
>0.5%: 54.3%
>1%: 41.6%
>2%: 24.3%
Exceedance probability
Heat-flow increase from nominal (%)
b
Severity controls the upper tail
bin median
bin 90th percentile
Heat-flow increase (%)
Bound on local gap deviation (%)
log count
c
Finite fields activate the narrow-gap branch
flat-conductance interval
Heat-flow increase (%)
Minimum local gap across both cavities (mm)
log count
d
Prescribed spatial-field envelope
Gap width (mm)
gap 1: mean and 5–95%
gap 2: mean and 5–95%
flat-interval bounds
nominal 10 mm
Position along section (mm)